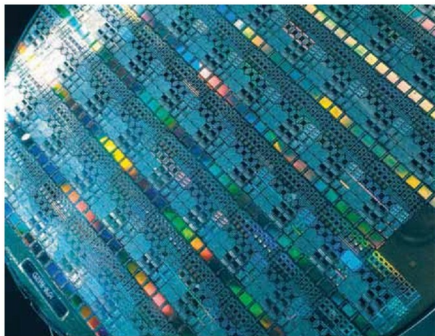


EMBED SUCCESS INTO YOUR CAREER!

Embedded systems offer a breadth of opportunity for their practitioners ranging from chip designing to system architecture and formulating testing strategy for software. So why not plan a career in embedded engineering?



■ MONIKA BHATI

Today, embedded systems are everywhere. Although you do not usually interact much with these systems, these tiny computers are present (besides PCs or workstations) in anything electronic that seems intelligent such as mobile phones, smart card readers, set-top boxes, microwave ovens, music systems, digital cameras, TVs, MP3 players, ATMs, automobiles, traffic signals and numerous other gadgets that we come across in our everyday life. Being a unique combination of computer hardware, software and sometimes additional mechanical

or other parts, embedded systems focus on one task and do it well. In other words, these sit inside your devices to add an element of smartness to them.

An embedded device is one in which the software is hidden in the hardware on which it runs. Embedded systems are usually designed to perform a specific function within a given time frame—for example, to help you choose for how long your washing machine should run, confer thinking power to the microwave oven and propel rocket launchers into space.

Let's figure out the opportunities in embedded industry and also if a career in this field could prove to be

a good take-off for you.

What's the market like?

Spurred by increasing sales of electronics and burgeoning telecom sector, the Indian semiconductor design industry is all set to reach \$10.2 billion by 2012, from \$7.5 billion in 2010, says a report by the India Semiconductor Association (ISA).

The Indian chip design industry, which comprises very-large-scale integration (VLSI) design, embedded software development and board design, is expected to grow 17.3 per cent year-on-year to reach the whopping figure of \$10.2 billion by 2012. This tremendous rate of growth will also require a large number of skilled professionals to increase the quality of work churned out.

The report further adds that the Indian semiconductor design industry employed a workforce of 160,000 in 2010, of which embedded software accounts for as much as 82 per cent employment. Thus highlighting the fact that embedded systems open up a plethora of opportunities for their practitioners.

India's semiconductor consumption is projected to reach \$8.2 billion in 2011, a 15.5 per cent jump from 2010 consumption of \$7.1 billion, according to research firm Gartner Inc. Based on this forecast, India is the fastest growing market in terms of semiconductor consumption for 2011.

"Changing demographics, increasing consumer affluence, economic growth and favourable government